PENETRATION TESTING REPORT

FOOTPRINTING & NETWORK SCANNING PHASES

W2-PM-FINAL | CYBERSECURITY | NETWORKWALKS

Report	W2-PM-FINAL | Cybersecurity | NetworkWalks
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Modules	W2-PM1 – Footprinting; W2-PM5 – Network Scanning
Targets	networkwalks.com (authorized educational scope); user's own local LAN
Environment	Kali Linux; Zenmap/Nmap on Kali Linux VM using bridged networking

Prepared for educational assessment. Findings are limited to the scope and evidence documented in this report.

1. Liability Disclaimer

This report documents educational reconnaissance and network-discovery activities. Testing of networkwalks.com was performed as part of the assigned NetworkWalks program and within the authorized scope represented by the assignment. Local network discovery was limited to the user's own LAN and lab devices. No exploitation, credential attacks, denial-of-service activity, or attempts to gain unauthorized access were performed. These findings are observations from the tests conducted and should not be treated as proof of exploitable vulnerabilities. Any further testing must have explicit authorization and a clearly defined scope.

2. Introduction

This report covers two Week 2 practical modules: W2-PM1 (Footprinting) and W2-PM5 (Network Scanning). W2-PM1 used Kali Linux tools to collect publicly available domain, DNS, web-technology, HTTP-header, and WAF information about networkwalks.com. W2-PM5 used Zenmap/Nmap to discover active devices on the user's own local network, identify IP and MAC addresses, and produce a network topology.

The work demonstrates two early stages of a security assessment: gathering information about an in-scope public-facing domain and identifying devices on an owned/local network. The report records the actual observed results, including limitations and tool errors. The Zenmap work was performed inside a Kali Linux virtual machine configured with a bridged network adapter; it was not performed on a Windows PC as described in the sample report.

3. Scope, Authorization and Environment

W2-PM1 target: https://networkwalks.com. The work was limited to passive or low-impact information gathering and basic web fingerprinting/DNS queries. W2-PM5 target: the user's local LAN, 10.218.122.0/24, including the Kali VM, Windows host and default gateway identified during the exercise.

Environment: Kali Linux VM; Nmap 7.99; WAFW00F 2.4.2; DNSRecon; DIG; WHOIS; WhatWeb; nslookup; curl; Zenmap GUI. Kali's bridged interface eth1 had address 10.218.122.138/24 and MAC address 08:00:27:8E:B4:19. Default gateway: 10.218.122.149.

4. Tools Used

Tool	Purpose
WHOIS	Retrieve public domain-registration details and registered name-server information.
DIG	Query DNS A, NS, MX, TXT and SOA records.
WhatWeb	Identify technologies and web application components exposed by the site.
nslookup	Resolve the domain and inspect A/MX/name-server information.
curl -I	Inspect HTTP response headers without downloading the full page.
WAFW00F	Fingerprint whether a web application firewall is detected.
DNSRecon	Enumerate DNS records and related service records.
Nmap / Zenmap	Discover live hosts on the local subnet and perform targeted host discovery.
ip link	Read the MAC address assigned to the Kali network interface.

5. Activities Performed and Findings

5.1 W2-PM1 – Footprinting and Reconnaissance

WHOIS: The saved WHOIS output identifies GoDaddy.com, LLC as registrar. It reports a domain creation date of 6 November 2019 and registry expiration date of 6 November 2027. The registrant details are privacy-protected through Domains By Proxy, LLC. The output lists HostGator name servers ns6135.hostgator.com and ns6136.hostgator.com. WHOIS reports DNSSEC as unsigned.

DIG and DNS lookups: networkwalks.com resolved to IPv4 address 192.232.216.135. The NS records returned ns6135.hostgator.com and ns6136.hostgator.com. The MX record returned priority 0 for mail.networkwalks.com, which also resolved to 192.232.216.135. TXT records included a Google site-verification value and an SPF policy: v=spf1 +a +mx +ip4:50.87.144.87 +include:websitewelcome.com ~all.

WhatWeb: The target returned HTTP 200 OK. The output identified Apache, WordPress 7.1.4, WordPress Download Manager 3.3.76, jQuery 3.7.1, Bootstrap 7.1.4, and the page title Networkwalks Academy. The detected server IP was 192.232.216.135. These are technology-fingerprinting observations, not confirmation that any component is vulnerable.

nslookup: The A lookup returned 192.232.216.135. The MX lookup returned mail.networkwalks.com with priority 0 and showed the two HostGator name servers.

curl -I: The saved HTTP response was HTTP/1.1 200 OK and identified Apache as the server. Response headers included a WordPress REST API link at /wp-json/ and a page endpoint under /wp-json/wp/v2/pages/53. The output also included security/privacy-related headers such as Permissions-Policy and Referrer-Policy.

WAFW00F: The corrected scan against https://networkwalks.com using WAFW00F v2.4.2 reported: “The site https://networkwalks.com is behind ModSecurity (SpiderLabs) WAF.” The tool reported two requests. This is a tool fingerprint result and does not establish the effectiveness or completeness of the WAF configuration.

DNSRecon: DNSRecon completed enumeration and reported eight SRV records for _autodiscover._tcp.networkwalks.com, targeting cpanelemaildiscovery.cpanel.net on port 443. It also returned SOA, NS, MX, A and TXT information. It identified the DNS server software banner as BIND 9.16.23-RH on the two name-server IPs it queried. The DNSSEC query produced “No answer for DNSSEC query”; this is recorded as a query observation. The WHOIS result separately reported DNSSEC unsigned.

SOA: The SOA response identified ns6135.hostgator.com as the primary name server and root.gator3068.hostgator.com as the responsible mailbox field. Serial: 2026111200; refresh: 86400 seconds; retry: 7200 seconds; expire: 3600000 seconds; minimum: 86400 seconds.

5.2 W2-PM5 – Network Scanning with Zenmap/Nmap

The Kali VM was changed to bridged networking so it could participate in the same LAN as the Windows host. The Kali interface eth1 received 10.218.122.138/24. The subnet used for discovery was 10.218.122.0/24 (subnet mask 255.255.255.0).

A subnet ping scan using Nmap host discovery (-sn) scanned 256 addresses and reported two hosts up: 10.218.122.1 and 10.218.122.138. The Windows host at 10.218.122.1 had MAC address 2C:3B:70:F8:7F:B1 (identified by Nmap as AzureWave Technology). The Kali VM's MAC address was 08:00:27:8E:B4:19, obtained from interface eth1 using ip link.

The default gateway 10.218.122.149 was checked separately with a targeted Nmap ping scan. It was up and returned MAC address 0A:26:3F:5F:06:CA (vendor unknown in the scan output). Therefore, three unique devices were identified across the full subnet scan and the separate gateway scan. The report does not claim that all three appeared in the same full-subnet scan.

The host inventory was saved in 17_host_inventory.txt. A network topology was saved as 14_network_topology.png and 14_network_topology.pdf. The practice-lab completion result was reported as 100%; the screenshot should be included as a separate evidence attachment if available.

6. Summary of Results

Area	Verified result
Domain IPv4	192.232.216.135
Name servers	ns6135.hostgator.com; ns6136.hostgator.com
Mail exchanger	mail.networkwalks.com (priority 0); resolves to 192.232.216.135
Web technologies	Apache; WordPress 7.1.4; WordPress Download Manager 3.3.76; jQuery 3.7.1; Bootstrap 7.1.4
WAF fingerprint	ModSecurity (SpiderLabs), detected by WAFW00F v2.4.2
DNSRecon	8 SRV records found; SOA/NS/MX/A/TXT information returned
LAN subnet scan	256 addresses scanned; 2 hosts up
Separate gateway scan	10.218.122.149 up; MAC 0A:26:3F:5F:06:CA
Unique LAN devices identified	3 (Windows host, Kali VM, default gateway)

7. Risk Analysis and Impact

The observations below indicate information that may assist an attacker during reconnaissance. They are not confirmed vulnerabilities. Risk levels are qualitative and provisional; no exploitability testing or vulnerability validation was performed.

Observation	Potential impact	Indicative level
Public technology fingerprinting (WordPress, plugin and library versions)	May help an attacker identify components for further research and security review.	Medium
Publicly resolvable server and mail IP address	Reveals hosting/network information that can support infrastructure profiling.	Low

Observation	Potential impact	Indicative level
HTTP headers and WordPress REST API paths disclosed	May provide additional application and technology details for further enumeration.	Low
WAF fingerprint identified as ModSecurity (SpiderLabs)	Reveals a security control and its product family; does not prove that it is misconfigured.	Low
DNS records and service records exposed	Can help map mail and DNS-related infrastructure.	Medium
Three unique devices identified on the user's LAN	Host discovery provides visibility into devices; unknown devices should be checked against an approved asset inventory.	Informational

No exploitation or vulnerability validation was performed. The localhost service-detection output in 10_service_detection.txt is a separate Kali-local practice scan (127.0.0.1), not a scan of networkwalks.com or the user's LAN, and is therefore not used to make claims about either target.

8. Recommendations

1. Review publicly exposed CMS, plugin and library information and keep WordPress, plugins, themes and JavaScript libraries updated.

2. Review HTTP response headers and WordPress REST API exposure; retain required functionality while avoiding unnecessary disclosure.

3. Maintain and monitor the detected ModSecurity WAF. Review its rules, logging, updates and false-positive handling; do not assume detection alone means the WAF is effective.

4. Review DNS records periodically. Remove obsolete records and ensure that mail, service and TXT records are accurate and necessary.

5. Confirm SPF, DKIM and DMARC configuration for the domain's email environment. The SPF TXT record was observed, but DKIM and DMARC were not tested in this exercise.

6. Maintain an approved LAN asset inventory and investigate devices that are not recognized. Repeat host discovery only within the authorized network scope.

7. Keep topology diagrams, IP/MAC inventories and scan evidence current, and store them securely.

8. Conduct any deeper vulnerability assessment only with written authorization, a defined scope, agreed testing windows and suitable safeguards.

9. Limitations and Notes

The subnet ping scan detected two hosts; the gateway was found by a separate targeted scan. These results are reported separately to avoid overstating the subnet scan result.

The Zenmap/Nmap scan was performed from Kali Linux in bridged mode, although the sample report describes Windows with Zenmap. This report documents the environment actually used.

The W2-PM1 Nmap service-detection file 10_service_detection.txt targets 127.0.0.1. It identified local SSH, Apache HTTP and a MariaDB banner. It is treated only as a local practice exercise and is not evidence about the external domain.

DNSRecon reported no answer for its DNSSEC query, while WHOIS reported DNSSEC unsigned. No additional DNSSEC validation was conducted.

No screenshots of every individual command output were available in the supplied evidence bundle. The saved text outputs, host inventory and topology files are listed in the evidence register below. Add the original topology and 100% practice-lab screenshots as separate attachments when submitting, if your portal allows multiple files.

10. Conclusion

The Week 2 practical work covered footprinting and network scanning. In W2-PM1, WHOIS, DIG, WhatWeb, nslookup, curl, WAFW00F and DNSRecon were used to collect domain-registration, DNS, web-technology, HTTP-header and WAF fingerprint information for networkwalks.com. The corrected WAFW00F scan detected ModSecurity (SpiderLabs).

In W2-PM5, Zenmap/Nmap host discovery was used on the user's own 10.218.122.0/24 LAN. The full subnet scan found two hosts, and a separate gateway scan found one additional host, giving three unique devices in the inventory. The topology was saved in PNG and PDF formats.

The exercises reinforced that accurate scope, careful interpretation and clear documentation are essential in cybersecurity. Reconnaissance results are leads for security review, not proof of vulnerabilities. All further testing must remain authorized and

within the agreed scope.

11. Evidence Register

Module	Evidence file	What it supports
W2-PM1	01_whois.txt	Domain registration, registrar, dates, name servers, DNSSEC status.
W2-PM1	02_dig.txt; 02_dns_records.txt; 08_dig_records.txt	A, NS, MX and TXT DNS query results.
W2-PM1	03_whatweb.txt	Web technology fingerprint.
W2-PM1	04_nslookup.txt	A and MX lookup results.
W2-PM1	05_curl.txt	HTTP response headers.
W2-PM1	06_wafw00f.txt	Correct WAFW00F result for networkwalks.com.
W2-PM1	07_dnsrecon.txt	DNS enumeration and SRV records.
W2-PM1	09_soa_record.txt	SOA record values.
W2-PM1	10_service_detection.txt	Kali localhost service-detection practice only.
W2-PM5	11_lan_ping_scan.txt; 13_mac_discovery.txt; 15_final_lan_scan.txt	Subnet host discovery; two hosts up.
W2-PM5	12_kali_mac.txt	Kali eth1 MAC address.
W2-PM5	16_gateway_check.txt	Separate gateway discovery and MAC address.
W2-PM5	17_host_inventory.txt	Consolidated three-device inventory.
W2-PM5	14_network_topology.png; 14_network_topology.pdf	Network topology export.
W2-PM5	Practice-lab completion screenshot	Reported 100% completion; attach screenshot separately if available.

Appendix A. Key Commands

```
whois networkwalks.com
dig networkwalks.com A
dig networkwalks.com NS
dig networkwalks.com MX
dig networkwalks.com TXT
dig networkwalks.com SOA
whatweb https://networkwalks.com
nslookup networkwalks.com
nslookup -type=MX networkwalks.com
curl -I https://networkwalks.com
wafw00f https://networkwalks.com
dnsrecon -d networkwalks.com
sudo nmap -sn 10.218.122.0/24
ip link show eth1
sudo nmap -sn 10.218.122.149
```

Appendix B. Final Host Inventory

Device	IP address	MAC address
Windows host	10.218.122.1	2C:3B:70:F8:7F:B1
Kali Linux VM (eth1)	10.218.122.138	08:00:27:8E:B4:19
Default gateway	10.218.122.149	0A:26:3F:5F:06:CA